

Assignment4 SU4: Implementing a Food Delivery Order Management System with the MyLinkedList class

Write a simple JAVA food delivery order management program that will use the provided MyLinkedList Class.

The program should display a menu with the following options:

- Place a new order
- Place an urgent order
- Process the next order
- Cancel an order
- Search for an order
- Display all current orders
- Display the first and last orders
- Clear all orders
- Exit

The information is maintained in a linked list of food delivery orders. Normal orders are added to the end of the list, while urgent orders are added to the beginning of the list. When an order is processed, the first order in the list is removed. The program must use the supplied MyLinkedList class **and must not use** Java's built-in LinkedList or other collection classes for storing the orders. The following functions should be implemented:

1. placeOrder() (12 marks)

This function should also allow the following:

- Collecting customer details: customer ID, name, contact number, and delivery address.
- Collecting order details: order ID, food item, quantity, price per item, and order type.
- Checking that the Order ID is unique before adding the order.
- Adding a normal order to the end of the linked list using append().
- Calculating and displaying the total order cost: quantity × price per item.
- Displaying a confirmation message after the order has been successfully added.

2. placeUrgentOrder() (8 marks)

This function should also allow the following:

- Collecting the same customer and order information required for a normal order.
- Checking that the Order ID is unique.
- Adding an urgent order to the beginning of the linked list using prepend().
- Displaying a confirmation that the urgent order has priority.
- Demonstrating that an urgent order is positioned before existing normal orders.

3. processNextOrder() (8 marks)

This function should also allow the following:

- Checking whether there are orders waiting to be processed.
- Displaying the first order before it is removed.
- Processing the first order in the linked list.
- Removing the processed order using removeFirst().
- Displaying a confirmation message after successful processing.

4. cancelOrder() (8 marks)

This function should also allow the following:

- Searching for an order using its Order ID.
- Displaying the order details when the order is found.
- Requesting confirmation before cancelling the order.
- Removing the order from the linked list using delete().
- Displaying an appropriate message when the order does not exist.

5. searchOrder() (6 marks)

This function should also allow the following:

- Searching by Order ID, Customer ID, customer name, or contact number.
 - Traversing the linked list to find a matching order.
 - Displaying all order information when a match is found.
 - Displaying an appropriate message when no matching order exists.
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6. displayOrders()**(5 marks)**

This function should also allow the following:

- Displaying all orders in their current linked-list order.
- Displaying Order ID, customer name, food item, quantity, total price, and order type.
- Using the supplied MyLinkedList functionality where appropriate.
- Displaying a suitable message when the list is empty.

7. displayFirstAndLastOrders()**(3 marks)**

- Use `getFirst()` to display the first order.
- Use `getLast()` to display the last order.
- Correctly handle an empty linked list.

8. clearOrders()**(3 marks)**

- Ask the user to confirm before removing all orders.
- Use `clear()` to remove all orders from the linked list.
- Display a confirmation message after the list has been cleared.

9. Order class**(5 marks)**

Create an appropriate class called `Order`. The class must contain:

- `orderId`, `customerID`, `customerName`, `contactNumber`, `deliveryAddress`.
- `foodItem`, `quantity`, `pricePerItem`, and `orderType`.
- A constructor and appropriate accessor/mutator methods.
- A method for calculating the total order cost.
- A meaningful `toString()` method.

10. Comparable<Order> implementation**(2 marks)**

- The `Order` class must implement `Comparable<Order>`.
- `compareTo()` must compare orders using the `Order ID`.

Additional requirements:

- The main order collection must be a `MyLinkedList<Order>`.
- Do not use `java.util.LinkedList`, `ArrayList`, `Vector`, `Set`, `Map`, or another collection class to store orders.
- Validate quantity, price, required text fields, order type, and duplicate `Order IDs`.
- Correctly handle an empty list for processing, searching, cancelling, displaying, and clearing.
- Submit `MyLinkedList.java`, `Order.java`, and `FoodDeliverySystem.java`.
- Test the program with at least 5 normal orders, 2 urgent orders, 1 cancellation, 1 processed order, an unsuccessful search, a duplicate ID attempt, and an empty-list scenario.